

1. Inference, statistics and estimators

Statistical inference uses data to learn about the parameters of a probability model. Following the lecture, π denotes a parameter or parameter vector. For independent and identically distributed (i.i.d.) observations,

$$\mathcal{X} = (X_1, \dots, X_n), \quad X_i \sim p(x | \pi).$$

In frequentist inference, π is **fixed but unknown**; the data vary between repeated experiments.

A **statistic** is a function $T(\mathcal{X})$ of the data that does not depend on unknown parameters. An **estimator** $\hat{\pi} = T(\mathcal{X})$ is a statistic used to estimate π . Before observing the data it is a random variable; after observing $\mathbf{x}$, $T(\mathbf{x})$ is a numerical **estimate**.

The **sampling distribution** $p(\hat{\pi} | \pi)$ describes the estimator over repeated datasets at the same true π . It determines the estimator's bias and scatter. It is not a distribution of the unknown parameter given the observed data.

An estimator can include an entire analysis pipeline or a neural network. Processing data cannot create information about π ; compression may preserve information or discard it.

Model correctness matters. With a misspecified model, more data can concentrate an estimate around the wrong value. Fisher precision under an assumed model does not test that model's validity.

2. Gaussian examples: mean and variance

Assume $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. The **sample mean** is

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Thus $\mathbb{E}[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$. The standard deviation of the estimator is $\sigma/\sqrt{n}$; the standard deviation of an individual observation is σ .

With known σ^2 , $\bar{X}$ is **sufficient for μ** : it retains all the information about μ in this Gaussian dataset. Sufficiency is a property relative to a specified model and parameter.

For $n > 1$, the **unbiased sample variance** is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Consequently,

$$\mathbb{E}[S^2] = \sigma^2, \quad \text{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

The $n-1$ denominator corrects for estimating the mean from the same data. Even though the observations are Gaussian, the sampling distribution of S^2 is a scaled chi-square distribution.

3. Bias, variance and mean-squared error

All expectations below average over repeated data at fixed π . For a scalar estimator,

$$\begin{aligned} b(\pi) &= \text{Bias}_\pi(\hat{\pi}) = \mathbb{E}_\pi[\hat{\pi}] - \pi, \\ \text{Var}_\pi(\hat{\pi}) &= \mathbb{E}_\pi[(\hat{\pi} - \mathbb{E}_\pi[\hat{\pi}])^2], \\ \text{MSE}_\pi(\hat{\pi}) &= \mathbb{E}_\pi[(\hat{\pi} - \pi)^2] \\ &= \text{Var}_\pi(\hat{\pi}) + b(\pi)^2. \end{aligned}$$

Unbiasedness means $b(\pi) = 0$ for every allowed π . It is an average property, not a guarantee that one estimate equals the truth. Variance describes scatter about the estimator's own mean.

For a parameter vector, write $\mathbf{b} = \mathbb{E}[\hat{\pi}] - \pi$ and $C = \text{Cov}(\hat{\pi})$. Squared Euclidean error gives

$$\mathbb{E}[\|\hat{\pi} - \pi\|^2] = \text{tr } C + \|\mathbf{b}\|^2.$$

Bias-variance tradeoff. A biased estimator can have a smaller MSE if its variance is sufficiently lower. For example, variance 1 and zero bias give MSE 1; variance 0.04 and bias 0.1 give MSE 0.05. The best estimator depends on the criterion being used.

Consistency means concentration on the truth as data accumulate:

$$\hat{\pi}_n \xrightarrow{P} \pi, \quad \text{i.e.} \quad \mathbb{P}_\pi(|\hat{\pi}_n - \pi| > \varepsilon) \rightarrow 0$$

for every $\varepsilon > 0$. Unbiasedness alone is insufficient: X_1 estimates a Gaussian mean without bias but does not improve as n grows; $\bar{X}_n$ is consistent. Bias may decrease with n , but vanishing bias alone does not imply consistency either.

4. Nonlinear transformations can introduce bias

Suppose $X_i \sim \mathcal{N}(f(\pi), \sigma^2)$ independently and use $\hat{\pi} = g(\bar{X})$. Assume f is smooth and one-to-one in a neighbourhood of the true parameter, with $f'(\pi) \neq 0$; $g = f^{-1}$ denotes its local inverse. Write $\bar{X} = f(\pi) + \epsilon$, with $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2/n$. A local expansion gives

$$\hat{\pi} \approx \pi + g'(f(\pi))\epsilon + \frac{1}{2}g''(f(\pi))\epsilon^2.$$

Using derivatives of the inverse function,

$$\begin{aligned} \text{Bias}_\pi(\hat{\pi}) &\approx -\frac{f''(\pi)}{2[f'(\pi)]^3} \frac{\sigma^2}{n}, \\ \text{Var}_\pi(\hat{\pi}) &\approx \frac{1}{[f'(\pi)]^2} \frac{\sigma^2}{n}. \end{aligned}$$

These small-noise or large- n approximations also require finite moments and controlled Taylor remainders. Local invertibility alone is insufficient. An unbiased estimate of $f(\pi)$ need not give an unbiased estimate of π after inversion. In general $\mathbb{E}[g(Z)] \neq g(\mathbb{E}[Z])$; Jensen's inequality gives the direction of this difference when g is convex or concave.

5. Likelihood and maximum-likelihood estimation

For an observed i.i.d. dataset $\mathbf{x} = (x_1, \dots, x_n)$,

$$\mathcal{L}(\pi; \mathbf{x}) = \prod_{i=1}^n p(x_i | \pi), \quad \ell(\pi) = \log \mathcal{L}(\pi; \mathbf{x}).$$

The **likelihood** varies π while holding the data fixed. It is not generally a normalised probability density over π . For dependent observations, use the joint data distribution instead of this product.

The **maximum-likelihood estimator (MLE)** is

$$\hat{\pi}_{\text{ML}} = \arg \max_{\pi} \ell(\pi).$$

At a differentiable interior maximum, $\nabla_{\pi} \ell(\hat{\pi}) = 0$. A stationary point must still be checked to be a maximum; boundary solutions need not have zero gradient.

Gaussian example ($n > 1$). If μ and σ^2 are unknown,

$$\hat{\mu}_{\text{ML}} = \bar{X}, \quad \hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_i (X_i - \bar{X})^2.$$

The variance MLE uses n , not $n-1$: $\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = (n-1)\sigma^2/n$. **Maximum likelihood does not guarantee finite-sample unbiasedness.**

6. Score, curvature and Fisher information

The **score** measures sensitivity of the log-likelihood to the parameters; the **observed information** measures its curvature:

$$s(\pi; \mathcal{X}) = \nabla_{\pi} \ell(\pi; \mathcal{X}), \quad J_n(\pi; \mathcal{X}) = -\nabla_{\pi}^2 \ell(\pi; \mathcal{X}).$$

Both can vary between datasets. Under regularity conditions, $\mathbb{E}_{\pi}[s(\pi; \mathcal{X})] = 0$ at a fixed generating parameter π . This expectation identity differs from the zero score at an interior MLE for one observed dataset.

The **Fisher information** in the full dataset is

$$\begin{aligned} F_n(\pi) &= \mathbb{E}_{\pi}[s(\pi; \mathcal{X})s(\pi; \mathcal{X})^{\top}] \\ &= \mathbb{E}_{\pi}[J_n(\pi; \mathcal{X})]. \end{aligned}$$

It is the covariance of the score, or equivalently the expected *negative* Hessian of the log-likelihood. These identities require conditions allowing differentiation under the probability integral; they are not automatic for every model.

For i.i.d. observations, information adds: $F_n = nF_1$. For a parameter vector, F_n is a symmetric positive semidefinite matrix. A larger eigenvalue indicates more local information along the corresponding parameter direction. Near an interior MLE, with $\delta\pi = \pi - \hat{\pi}$,

$$\ell(\pi) \approx \ell(\hat{\pi}) - \frac{1}{2} \delta\pi^{\top} J_n(\hat{\pi}; \mathbf{x}) \delta\pi.$$

Greater curvature means a narrower local likelihood. Observed information describes one dataset; Fisher information averages over hypothetical repeated datasets.

7. Gaussian Fisher information

Let each observation be a vector $\mathbf{X}_i \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$ independently, with known positive definite covariance Σ . When the parameter is the mean itself,

$$\begin{aligned} s(\boldsymbol{\mu}) &= n\Sigma^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}), \\ J_n(\boldsymbol{\mu}; \mathcal{X}) &= F_n = n\Sigma^{-1}, \\ \bar{\mathbf{X}} &\sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma/n). \end{aligned}$$

Thus $\text{Cov}(\bar{\mathbf{X}}) = \Sigma/n = F_n^{-1}$ exactly. For a scalar mean, $F_n = n/\sigma^2$ and the standard error is $1/\sqrt{F_n} = \sigma/\sqrt{n}$.

More generally, if the mean is $\boldsymbol{\mu}(\boldsymbol{\pi})$ and Σ is known and *independent of the parameters*, define $D_{\alpha\alpha} = \partial\mu_{\alpha}/\partial\pi_{\alpha}$. Then

$$F_n(\boldsymbol{\pi}) = nD^{\top}\Sigma^{-1}D.$$

Information increases when the predicted mean changes strongly relative to the noise. If Σ depends on the parameters, its derivatives contribute additional terms.

8. The Cramér–Rao bound and efficiency

For a **regular unbiased estimator**, with nonsingular Fisher information, the scalar and vector bounds are

$$\text{Var}_{\pi}(\hat{\pi}) \geq \frac{1}{F_n(\pi)}, \quad \text{Cov}_{\pi}(\hat{\boldsymbol{\pi}}) \succeq F_n^{-1}.$$

The matrix inequality means the difference is positive semidefinite: the variance bound applies to every fixed linear combination of the estimated parameters. It is a statement about repeated sampling at a fixed true parameter.

An unbiased estimator that attains the bound is **efficient**. The Gaussian sample mean above is an exact example. The bound depends on the model; actual covariance also depends on the estimator.

Scope of the bound. Biased estimators can have variance below F_n^{-1} without contradiction. The bound need not be attainable at finite n , and it does not specify how many observations make an approximation accurate. Nonlinear models can satisfy the bound's assumptions; nonlinearity alone does not invalidate it.

Under suitable regularity and identifiability conditions, MLEs are consistent and asymptotically efficient:

$$\sqrt{n}(\hat{\boldsymbol{\pi}}_{\text{ML}} - \boldsymbol{\pi}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, F_1(\boldsymbol{\pi})^{-1}).$$

This gives the large-sample approximation $\text{Cov}(\hat{\boldsymbol{\pi}}_{\text{ML}}) \approx F_n^{-1}$ under the corresponding moment conditions, not a universal finite-sample identity.

9. Choosing and checking an estimator

Sample means, medians and neural networks can differ in bias, variance, MSE and information retention. Check their sampling distributions by repeating the full analysis on simulated datasets across the parameter range of interest.

Interpreting uncertainty regions requires further frequentist or Bayesian constructions; a point estimate alone does not supply them.